

Combination, Relevance, and Comparability

A tacit assumption underlies our previous examples of simultaneous inference: that all cases presented together should be analyzed together, such as all 6033 genes for the prostate data in Figure 5.2, or all 15 443 DTI voxels. This leads us down a perilous path. Omnibus combination may distort individual inferences in both directions: interesting cases may be hidden while uninteresting ones are enhanced. This chapter concerns the separation and combination of cases in a large-scale data analysis.

Figure 10.1 illustrates separation/combination difficulties in terms of the DTI data. Z-values for the 15 443 voxels have been separated into back and front halves of the brain ($x < 50$ and $x \geq 50$ in terms of Figure 2.4), 7661 voxels in back and 7782 in front. Two discrepancies strike the eye: the heavy right tail in the combined histogram of Figure 5.1b comes exclusively from the front; and the center of the back-half histogram is shifted leftwards 0.35 units relative to the front.

Separate local false discovery rate analyses, assuming a theoretical null distribution $N(0, 1)$ as in Chapter 5, gave 271 voxels with $\widehat{\text{fdr}}(z_i) \leq 0.20$ for the front-half data, those having $z_i \geq 2.71$, but none for the back half. This is quite different from the combined analysis of Figure 5.2, where only the 184 voxels with $z_i \geq 3.05$ (including nine from the back) had $\widehat{\text{fdr}}(z_i) \leq 0.20$.

Scientific guidance would be most welcome at this point, but in its absence the statistical evidence of Figure 10.1 argues against a combined analysis. Section 10.2 and Section 10.3 go further, suggesting still finer separation of cases, while describing a convenient methodology for doing so. The basic methodological idea is developed in Section 10.1. Applications to enrichment are discussed in Section 10.2. Section 10.4 discusses whether false discovery rate methods maintain their inferential legitimacy in the face of separation. *Comparability*, Section 10.5, concerns the type of statistics that are appropriate for the combination of cases.

The central concern of this chapter is familiar in the family-wise error rate literature, where it takes the form of the question “What is a family?”

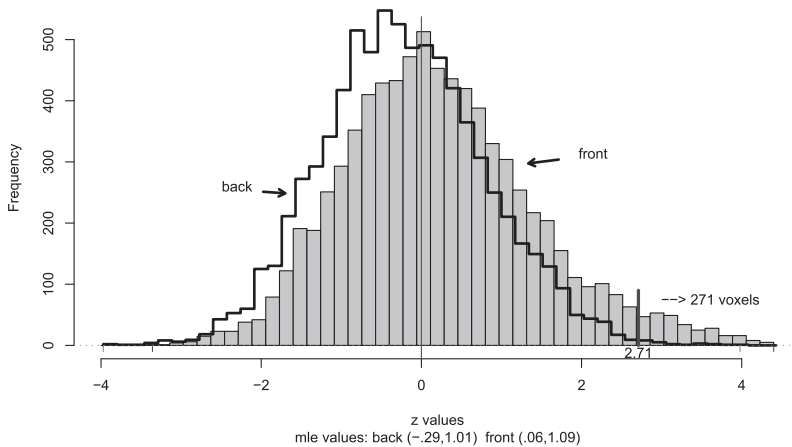


Figure 10.1 DTI data of Section 2.5 separated into back-half and front-half voxels ($x < 50$ and $x \geq 50$ in Figure 2.4). Separate analyses gave 271 voxels with $\widehat{\text{fdr}} \leq 0.20$ for front half, but none for back half. MLE empirical null estimates ($\hat{\delta}_0, \hat{\sigma}_0$) of Section 6.3 show back-half center shifted 0.35 units to the left of front half.

Miller's introduction to the second edition of his simultaneous inference book begins in this way:

Time has now run out. There is nowhere left for the author to go but to discuss just what constitutes a family. This is the hardest part of the entire book because it is where statistics takes leave of mathematics and must be guided by subjective judgement.¹

We are at an advantage today because large-scale data sets allow us, sometimes, to examine directly the trade-offs between separation and combination of cases. False discovery rates, this chapter's favored methodology, help clarify the question. Basically, however, we won't be able to answer Miller's question of what is the proper family for simultaneous analysis, though evidence like that in Figure 10.1 may strongly suggest what *is not*. Miller's "subjective judgements" still dominate practice. There are some hints in what follows of a more principled approach to separation and combination, particularly in the section on *relevance*, but at this stage they remain just hints.

¹ Section 1.5 of Westfall and Young's 1993 book gives another nice discussion of basic multiple inference conundrums.

10.1 The Multi-Class Model

The two-groups model of Section 2.2 and Section 5.1 can be extended to cover the situation where the N cases are divided into distinct classes, possibly having different choices of π_0 , $f_0(z)$, and $f_1(z)$ shown in Figure 2.3. Figure 10.2 illustrates the scheme: two classes A and B (e.g., the front and back halves of the brain in the DTI example) have a priori probabilities w_A and $w_B = 1 - w_A$. The two-groups model (5.1) holds separately within each class, for example with $\pi_0 = \pi_{A0}$, $f_0(z) = f_{A0}(z)$, and $f_1(z) = f_{A1}(z)$ in class A . It is important to note that the class label A or B is observed by the statistician, in contrast to the null/non-null dichotomy which must be inferred.

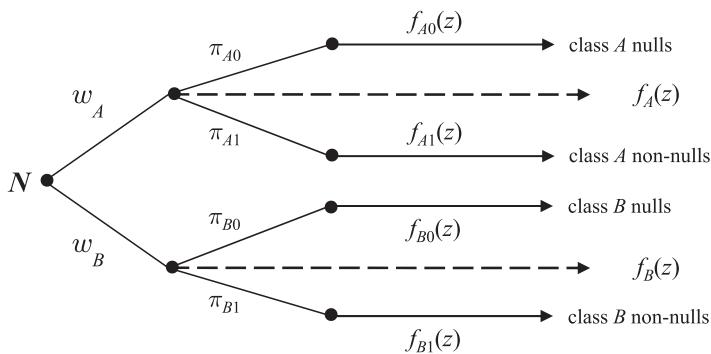


Figure 10.2 The multi-class model with two classes, A and B . Two-groups model (5.1) holds separately within each class, with possibly different parameters. Prior probabilities w_A and w_B shown for the two classes.

The definitions of Section 5.1 apply separately within classes, for instance to the class A mixture density and local false discovery rate

$$f_A(z) = \pi_{A0}f_{A0}(z) + \pi_{A1}f_{A1}(z) \quad \text{and} \quad \text{fdr}_A(z) = \pi_{A0}f_{A0}(z)/f_A(z). \quad (10.1)$$

A combined analysis, ignoring the class information in Figure 10.1, has marginal densities

$$\begin{aligned} f_0(z) &= w_A \frac{\pi_{A0}}{\pi_0} f_{A0}(z) + w_B \frac{\pi_{B0}}{\pi_0} f_{B0}(z) \\ f_1(z) &= w_A \frac{\pi_{A1}}{\pi_1} f_{A1}(z) + w_B \frac{\pi_{B1}}{\pi_1} f_{B1}(z) \end{aligned} \quad (10.2)$$

where $\pi_0 = 1 - \pi_1$ is the combined null probability,

$$\pi_0 = w_A \pi_{A0} + w_B \pi_{B0}. \quad (10.3)$$

This leads to overall marginal density and false discovery rate

$$f(z) = w_A f_A(z) + w_B f_B(z) \quad \text{and} \quad \text{fdr}(z) = \pi_0 f_0(z) / f(z) \quad (10.4)$$

just as in (5.2), (5.3). If class information were unavailable, we would have to base our inference on $\text{fdr}(z)$, but here we have the option of separately employing $\text{fdr}_A(z)$ or $\text{fdr}_B(z)$.

Bayes theorem yields a useful relationship between the separate and combined false discovery rates.

Theorem 10.1 *Define $w_A(z)$ as the conditional probability of a case being in class A given z ,*

$$w_A(z) = \Pr\{A|z\} \quad (10.5)$$

and similarly define

$$w_{A0}(z) = \Pr_0\{A|z\} \quad (10.6)$$

to be the conditional probability of class A given z for a null case. Then

$$\text{fdr}_A(z) = \text{fdr}(z) R_A(z) \quad \text{where } R_A(z) = \frac{w_{A0}(z)}{w_A(z)}. \quad (10.7)$$

Proof Let I be the event that a case is null, so I occurs in the two null paths of Figure 10.2 but not otherwise. We have

$$\begin{aligned} \frac{\text{fdr}_A(z)}{\text{fdr}(z)} &= \frac{\Pr\{I|A, z\}}{\Pr\{I|z\}} = \frac{\Pr\{I, A|z\}}{\Pr\{A|z\} \Pr\{I|z\}} \\ &= \frac{\Pr\{A|I, z\}}{\Pr\{A|z\}} = \frac{w_{A0}(z)}{w_A(z)}. \end{aligned} \quad (10.8) \quad \square$$

Tail-area false discovery rates also obey the theorem. In the general notation (2.12), (2.13),

$$\text{Fdr}_A(\mathcal{Z}) = \text{Fdr}(\mathcal{Z}) R_A(\mathcal{Z}) \quad \text{where } R_A(\mathcal{Z}) = \frac{\Pr_0\{A|z \in \mathcal{Z}\}}{\Pr\{A|z \in \mathcal{Z}\}}. \quad (10.9)$$

Exercise 10.1 Verify (10.9).

The virtue of Theorem 10.1 is that $R_A(z)$ is often easy to estimate, which allows $\widehat{\text{fdr}}_A(z)$ to be obtained from $\widehat{\text{fdr}}(z)$ even if class A is too small for direct estimation of fdr_A (see Section 10.2). We do not need the theorem

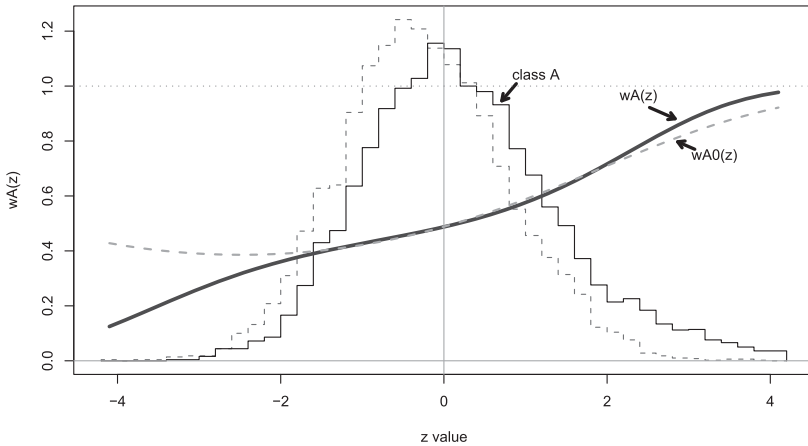


Figure 10.3 Heavy curve is logistic regression estimate of (10.5), $w_A(z) = \Pr\{A|z\}$, where class A is the front-half portion of the DTI data; dashed curve estimates $w_{A0}(z) = \Pr_0\{A|z\}$ (10.6).

for the DTI example of Figure 10.1 since class A is large, but the calculation of $R_A(z) = w_{A0}(z)/w_A(z)$ is interesting in its own right.

The heavy curve in Figure 10.3 is an estimate of $w_A(z) = \Pr\{A|z\}$ obtained by ordinary logistic regression: the class indicators

$$c_i = \begin{cases} 1 & \text{if case}_i \text{ in class A} \\ 0 & \text{if case}_i \text{ in class B} \end{cases} \quad (10.10)$$

were regressed as a cubic logistic function of z_i . As might be expected from the two histograms, $\hat{w}_A(z)$ increases smoothly with z and, in fact, a linear logistic regression would have sufficed here.

In order to estimate $w_{A0}(z) = \Pr_0\{A|z\}$, we need to make some assumptions about the null distributions in Figure 10.2. Assuming normal nulls and using the MLE method of Section 6.3 gave estimates

$$f_{A0}(z) \sim \mathcal{N}(\delta_{A0}, \sigma_{A0}^2) \quad \text{and} \quad f_{B0}(z) \sim \mathcal{N}(\delta_{B0}, \sigma_{B0}^2) \quad (10.11)$$

with

$$(\delta_{A0}, \sigma_{A0}) = (0.06, 1.09) \quad \text{and} \quad (\delta_{B0}, \sigma_{B0}) = (-0.29, 1.01). \quad (10.12)$$

The null probabilities were $\pi_{A0} = 0.969$ and $\pi_{B0} = 0.998$. The dashed curve in Figure 10.3 is $\hat{w}_{A0}(z)$ estimated, using Bayes rule, from (10.11)–(10.12).

Note: We assume $w_A = w_B = 0.50$ in Figure 10.2 since the front- and back-half sets are nearly equal in size.

Exercise 10.2 (a) Show that (10.11) gives

$$\frac{w_{A0}(z)}{w_{B0}(z)} = \frac{w_A \pi_{A0} \sigma_{B0}}{w_B \pi_{B0} \sigma_{A0}} \exp - \frac{1}{2} \left\{ \left(\frac{z - \delta_{A0}}{\sigma_{A0}} \right)^2 - \left(\frac{z - \delta_{B0}}{\sigma_{B0}} \right)^2 \right\}. \quad (10.13)$$

(b) Express $w_{A0}(z)$ in terms of $w_{A0}(z)/w_{B0}(z)$.

Looking at Figure 10.1, we might expect that $\widehat{\text{fdr}}_A(z)$ would be much lower than the combined estimate $\widehat{\text{fdr}}(z)$ for large values of z , but that is not the case. The empirical Bayes ratio

$$\hat{R}_A(z) = \hat{w}_{A0}(z)/\hat{w}_A(z) \quad (10.14)$$

is ≥ 0.94 for $z > 0$, so formula (10.7) implies only small differences. Two contradictory effects are at work: the longer right tail of the front-half distribution by itself would produce small values of $\hat{R}_A(z)$ and $\widehat{\text{fdr}}_A(z)$. However, the effect is mostly canceled by the rightward shift of the whole front-half distribution, which substantially increases the numerator of $\widehat{\text{fdr}}_A(z) = \hat{\pi}_{A0} \hat{f}_{A0}(z)/\hat{f}_A(z)$.

The close match between $\hat{w}_{A0}(z)$ and $\hat{w}_A(z)$ near $z = 0$ is no accident. Following through the definitions in Figure 10.3 and (10.1) gives, after a little rearrangement,

$$\frac{w_A(z)}{1 - w_A(z)} = \frac{w_{A0}(z)}{1 - w_{A0}(z)} \frac{1 + Q_A(z)}{1 + Q_B(z)} \quad (10.15)$$

where

$$Q_A(z) = \frac{1 - \text{fdr}_A(z)}{\text{fdr}_A(z)} \quad \text{and} \quad Q_B(z) = \frac{1 - \text{fdr}_B(z)}{\text{fdr}_B(z)}. \quad (10.16)$$

Often $\text{fdr}_A(z)$ and $\text{fdr}_B(z)$ approximately equal 1.0 near $z = 0$, reflecting a large preponderance of null cases and the fact that non-null cases tend to produce z -values farther away from zero. Then (10.15) gives

$$w_A(z) \doteq w_{A0}(z) \quad \text{for } z \text{ near zero} \quad (10.17)$$

as seen in Figure 10.3.

Suppose we believe that $f_{A0}(z) = f_{B0}(z)$ in Figure 10.2, i.e., that the null cases are distributed identically in the two classes. (This being true in particular if we accept the usual $\mathcal{N}(0, 1)$ theoretical null distribution.) Then $w_{A0}(z)$ does not depend on z , and we obtain a simplified version of Theorem 10.1.

Corollary 10.2 *If*²

$$f_{A0}(z) = f_{B0}(z) \quad (10.18)$$

for all z then

$$\text{fdr}_A(z) = c_0 \text{fdr}(z)/w_A(z) \quad [c_0 = w_A\pi_{A0}/\pi_0]. \quad (10.19)$$

Substituting $z = 0$ in (10.19) gives $c_0 = w_A(0) \text{fdr}_A(0)/\text{fdr}(0)$ and

$$\text{fdr}_A(z) = \left(\frac{\text{fdr}_A(0)}{\text{fdr}(0)} \right) \text{fdr}(z) \cdot \frac{w_A(0)}{w_A(z)}. \quad (10.20)$$

The first factor is often near 1, as mentioned before (10.17), yielding the convenient approximation

$$\text{fdr}_A(z) \doteq \text{fdr}(z) \frac{w_A(0)}{w_A(z)}. \quad (10.21)$$

(The missing factor $\text{fdr}_A(0)/\text{fdr}(0)$ in (10.21) is likely to be ≤ 1 if A is an enriched class, since “enrichment” implies lower null probabilities, making approximation (10.21) conservative.) Formula (5.25) can be shown to be a special case of (10.19).

Exercise 10.3 (a) Verify that Corollary 10.2 also applies to $\text{Fdr}_A(\mathcal{Z})$ (10.9) in the form

$$\text{Fdr}_A(\mathcal{Z}) = c_0 \text{Fdr}(\mathcal{Z})/w_A(\mathcal{Z}) \quad (10.22)$$

where $w_A(\mathcal{Z}) = \Pr\{A|z \in \mathcal{Z}\}$ and $c_0 = w_A\pi_{A0}/\pi_0$. (b) Show that (10.17) and (10.18) together imply

$$\left. \frac{dw_A(z)}{dz} \right|_{z=0} \doteq 0. \quad (10.23)$$

Approximation (10.23) can serve as a check on assumption (10.18). For instance, the solid curve in Figure 10.3 has $d\hat{w}_A(z)/dz$ noticeably positive near $z = 0$, casting doubt on (10.18) without recourse to normal models (10.11). This kind of argument is less convincing for small “enriched” classes A where $\text{fdr}_A(0)$ may be a lot less than 1, invalidating the reasoning leading up to (10.17).

Summary Making N , the number of cases, as large as possible is the rationale for combined analysis, at least in an empirical Bayes framework. Large N makes estimates like $\overline{\text{Fdr}}(\mathcal{Z})$ better approximations to the true

² Notice that (10.18) is equivalent to $f_{A0}(z) = f_0(z)$, using (10.2).

Bayes value $\text{Fdr}(\mathcal{Z})$, as in Section 2.4, justifying Bayesian interpretation of our results. The danger of combination is that we may be getting an accurate estimate of the wrong quantity: if $\text{Fdr}_A(z)$ is much different than $\text{Fdr}_B(z)$, then $\overline{\text{Fdr}}(z)$ may be far from both. Theorem 10.1 and Corollary 10.2 offer the possibility of accurately estimating $\text{Fdr}_A(z)$ even when A is a small subclass of all N cases.

10.2 Small Subclasses and Enrichment

Corollary 10.2 is especially useful when A is a *small* subclass, for which direct estimates $\widehat{\text{fdr}}_A(z)$ or $\widehat{\text{Fdr}}_A(z)$ would be hopelessly inaccurate because of inadequate sample size N . We encountered small subclasses in the p53 enrichment example of Chapter 9, and enrichment will be taken up later in this section, but smallness can force itself upon our attention even in the absence of preselected enrichment classes.

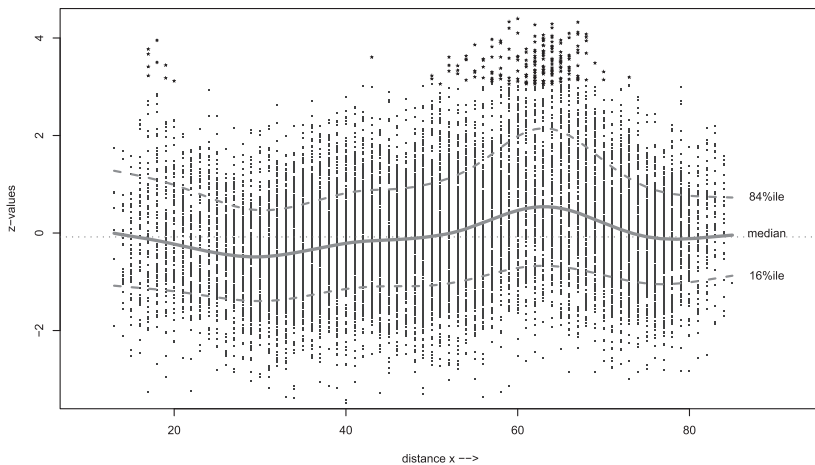


Figure 10.4 DTI z -values plotted versus distance x from back of brain; stars indicate the 184 voxels with $z_i \geq 3.05$, those having $\widehat{\text{fdr}}(z_i) \leq 0.20$ in combined analyses of all 15 443 voxels, Figure 5.2. Running percentile curves show a general upward shift of z -values around $x = 62$, the region including most of the starred voxels.

The $N = 15\,443$ DTI z -values z_i are plotted versus x_i , the voxel's distance from the back of the brain, in Figure 10.4. A clear wave is visible, cresting

near $x = 62$. Most of the 184 voxels identified as non-null in the combined $\widehat{\text{fdr}}$ analysis, those with $\widehat{\text{fdr}}(z_i) \leq 0.20$, occurred at the top of the crest.

There is something worrisome here: the z -values around $x = 62$ are shifted upward across their entire range, not just in the upper percentiles. This might be due to a reading bias in the DTI imaging device, or a genuine difference between dyslexic and normal children for *all* brain locations near $x = 62$. In neither case would it be correct to assign some special significance to those voxels near $x = 62$ that happen to have large z_i values.

Figure 10.4 suggests using finer subdivisions than the front/back split in Figure 10.1. As an example, let class A be the 82 voxels located at $x = 18$. These display some large z -values, attained without the benefit of riding a wave crest.³ Their histogram is compared with that of all other voxels in Figure 10.5, along with an estimate of the factor

$$R(z) = w_A(0)/w_A(z) \quad (10.24)$$

in Corollary 10.2, formula (10.21).

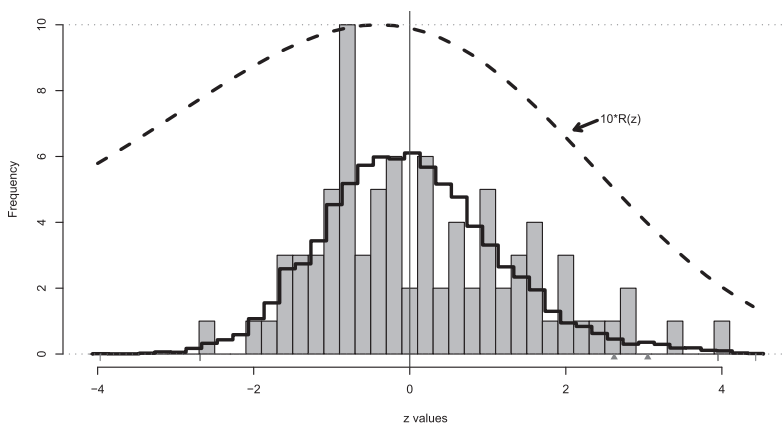


Figure 10.5 DTI z -values for the 82 voxels at $x = 18$ (solid histogram) compared with all other voxels (line histogram). Dashed curve is 10 times $\hat{R}(z) = \hat{w}_A(0)/\hat{w}_A(z)$, formula (10.21), Corollary 10.2; $\hat{R}(z)$ is less than 1 for large z , substantially reducing $\widehat{\text{fdr}}_A(z)$ below $\widehat{\text{fdr}}(z)$. Triangles indicate threshold values $\widehat{\text{fdr}}_A(z) = 0.20$ and $\widehat{\text{fdr}}(z) = 0.20$, at 2.62 and 3.05, respectively.

$\hat{R}(z)$ decreases for large values of z , making $\widehat{\text{fdr}}_A(z)$ substantially smaller

³ Figure 9.7b also suggests something interesting going on near $x = 18$.

than $\widehat{\text{fdr}}(z)$, the combined empirical null fdr estimate. The threshold value of z , that giving 0.20 as the estimated fdr, decreases from 3.05 for the combined estimate $\widehat{\text{fdr}}$ to 2.62 for $\widehat{\text{fdr}}_A$. The number of voxels having estimated fdr less than 0.20 increases from two to five. (Note that $\hat{R}(z) = \hat{w}_A(0)/\hat{w}_A(z)$ in (10.21) does not depend on how $\text{fdr}(z)$ is estimated. The dashed curve in Figure 10.5 applies just as well to empirical null estimation of $\text{fdr}(z)$ instead of theoretical null, in which case the threshold values increase to 2.68 and 3.17, respectively.)

Enrichment

Returning to the enrichment calculations of Chapter 9, we can reasonably state the null hypothesis that class A is *not* enriched as $\text{fdr}_A(z) = \text{fdr}(z)$ for all z , where $\text{fdr}(z)$ is the false discovery rate ignoring class (10.4); in other words, knowing that a case is in class A does not change our assessment of its null probability.

Assuming $f_{A0}(z) = f_0(z)$, as in Corollary 10.2, approximation (10.21) yields the

$$\text{enrichment null hypothesis: } w_A(z) = \text{constant} \quad (10.25)$$

which means we can use $\hat{w}_A(z)$ to test for enrichment. For instance, we might estimate $w_A(z)$ by linear logistic regression and then test (10.25) with

$$S = \hat{\beta}/\widehat{\text{se}} \quad (10.26)$$

where $\hat{\beta}$ is the regression slope and $\widehat{\text{se}}$ its standard error.

Slope statistic (10.26) was computed for the P53_UP pathway (the top row in Figure 9.5) comprising 40 of the 10 100 p53 data z -values: a standard linear logistic model for $w_A(z)$ gave $\hat{\beta} = 0.699 \pm 0.153$ and

$$S = 0.699/0.153 = 4.58, \quad (10.27)$$

two-sided p -value $4.7 \cdot 10^{-6}$. Permutation analysis gave $\hat{\beta} = 0.699 \pm 0.240$ and the more realistic value

$$S = 0.699/0.240 = 2.91, \quad (10.28)$$

$p = 0.0036$.

Tests based on (10.26) can be shown to match asymptotically those based on $S = \bar{z}_A$, the average of the z -values within A . (The connection is through model (9.38).) There is no particular advantage to testing enrichment with (10.26) rather than the methods of Chapter 9. However, having found a significantly enriched class A like P53_UP in Figure 9.5, we can use Corollary 10.2 to identify individual non-null cases *within* A . Formula

(10.21), with $\hat{w}_A(z)$ obtained from the linear logistic model above, yielded nine genes in P53_UP with $\widehat{\text{fdr}}_A(z_i) \leq 0.20$, compared to just one based on $\widehat{\text{fdr}}(z_i)$.

Exercise 10.4 Assuming $f_{A0}(z) = f_0(z)$ for all z , show that the null hypothesis $H_{00} : \text{fdr}_A(z) = \text{fdr}(z)$ for all z implies $H_0 : w_A(z) = \text{constant}$, but H_0 only implies $\text{fdr}_A(z) = \text{constant} \cdot \text{fdr}(z)$. (So rejection of H_0 implies rejection of H_{00} .)

Efficiency

How efficient is the formula based on Corollary 10.2,

$$\widehat{\text{fdr}}_A(z) = c_0 \widehat{\text{fdr}}(z) / \hat{w}_A(z) \quad (10.29)$$

as an estimate of $\text{fdr}_A(z)$? The following heuristic calculations suggest substantial, and sometimes enormous, savings over direct estimation, when A is small.

Taking logarithms in (10.29) gives

$$\widehat{\text{lfd}}_A(z) = \widehat{\text{lfd}}(z) - \widehat{\text{lw}}_A(z) + \text{constant} \quad (10.30)$$

with $\widehat{\text{lfd}}_A(z) = \log(\widehat{\text{fdr}}_A(z))$, etc. For reasons discussed later, $\widehat{\text{lfd}}_A(z)$ and $\widehat{\text{lw}}_A(z)$ are nearly uncorrelated, leading to a convenient approximation for the standard deviation of $\widehat{\text{lfd}}_A(z)$,

$$\text{sd}\{\widehat{\text{lfd}}_A(z)\} \doteq \left[\text{sd}\{\widehat{\text{lfd}}(z)\}^2 + \text{sd}\{\widehat{\text{lw}}_A(z)\}^2 \right]^{\frac{1}{2}}. \quad (10.31)$$

Of course, we expect $\widehat{\text{fdr}}_A(z)$ to be more variable than $\widehat{\text{fdr}}(z)$ since it is based on fewer cases, N_{w_A} rather than N . Standard “square root of sample size” considerations suggest

$$\text{sd}\{\widehat{\text{lfd}}_A(z)\} \sim \frac{1}{\sqrt{w_A}} \text{sd}\{\widehat{\text{lfd}}(z)\} \quad (10.32)$$

if $\text{fdr}_A(z)$ were estimated directly in the same manner as $\text{fdr}(z)$. The efficiency question is how does (10.31) compare with (10.32)?

Table 10.1 makes the comparison for a version of situation (5.18): independent z_i are generated as in Figure 10.2, with

$$\begin{aligned} w_A &= 0.01, & \pi_{A0} &= 0.50, & \pi_0 &= 0.95, \\ f_{A0} &= f_{B0} \sim \mathcal{N}(0, 1) & \text{and} & & f_{A1} &= f_{B1} \sim \mathcal{N}(2.5, 1), \end{aligned} \quad (10.33)$$

$N = 3000$ and 6000 . So for $N = 3000$, A had 30 members, half of them of the non-null $\mathcal{N}(2.5, 1)$ variety, compared with 5% non-null overall.

Table 10.1 *Ratio $\text{sd}\{\widehat{\text{lfdr}}_A(z)\} / \text{sd}\{\widehat{\text{lfdr}}(z)\}$ using formula (10.31) with linear logistic estimator $\widehat{w}_A(z)$; simulations from model (10.33). Bottom row is true $\text{fdr}(z)$.*

z	2	2.5	3	3.5	4	4.5
$N = 3000$	2.2	2.0	2.0	2.0	1.8	1.4
$N = 6000$	2.9	3.3	2.7	2.4	2.0	1.5
$\text{fdr}(z)$	.74	.46	.19	.066	.019	.006

The table shows values of $\text{sd}\{\widehat{\text{lfdr}}_A(z)\} / \text{sd}\{\widehat{\text{lfdr}}(z)\}$ of about 1.5 to 3.0, based on (10.31), in the range where the true $\text{fdr}(z)$ value is less than 0.50. This compares with a ratio of 10 suggested by (10.32). Here $\widehat{\text{fdr}}(z) = \pi_0 f_0(z) / \hat{f}(z)$ was calculated assuming $\pi_0 f_0(z)$ known. Empirical null estimation of $\pi_0 f_0(z)$ sharply reduces the ratios, to near 1 for large z , because $\text{sd}\{\widehat{\text{lfdr}}(z)\}$ increases (as in Table 7.3) while $\text{sd}\{\widehat{\text{lw}}_A(z)\}$ stays the same. In practice, direct empirical null estimation is impossible for a class A of only 60 members, leaving Corollary 10.2 as the sole hope for assessing $\widehat{\text{fdr}}_A(z)$.

The standard deviation approximation (10.31) depends on $\widehat{\text{lfdr}}(z)$ and $\widehat{\text{lw}}_A(z)$ being nearly uncorrelated. Let $\mathbf{y}$ be the vector of discretized counts for all N z_i values, as in Section 5.2, and $\mathbf{y}_A$ the counts for the z -values in A . The Poisson regression estimate $\widehat{\text{fdr}}$ depends on $\mathbf{y}$ while $\widehat{\text{lw}}_A$ depends on the ratios $\mathbf{r} = (\dots y_{Ak}/y_k \dots)$ with $\mathbf{y}$ itself entering only in an ancillary role. In outline, the argument for (10.31) depends on the general equality

$$\text{var}\{X + Y\} = \text{var}\{X\} + \text{var}\{Y\} + 2 \text{cov}\{X, E(Y|X)\} \quad (10.34)$$

applied to $X = \widehat{\text{lfdr}}_k$ and $Y = -\widehat{\text{lw}}_{Ak}$. Both $\text{var}\{X\}$ and $\text{var}\{Y\}$ are $O(1/N)$, but because the expectation of $\mathbf{r}$ does not depend on $\mathbf{y}$, the covariance term in (10.31) is only $O(1/N^2)$.

Exercise 10.5 Verify (10.34).

10.3 Relevance

So far we have discussed the separation of cases into two classes A and B (or A and not- A), of possibly distinct behavior. The previous theory can be extended to include multiple classes, say one for every value of x in Figure 10.4, incorporating a notion of relevance between the classes. In the

DTI example, for instance, we might assess the relevance between voxels at x_i and x_j by some smooth kernel such as $\exp\{-|x_i - x_j|/5\}$ rather than whether they occur in the same half of the brain.

Each case is now assumed to have three components,

$$\text{case } i = (x_i, z_i, I_i) \quad i = 1, 2, \dots, N, \quad (10.35)$$

where x_i is an observed vector of covariates, z_i is the observed z -value, and I_i is an unobservable null indicator, having I_i either 1 or 0 as case i is null or non-null. The two-groups model (5.1) is assumed to hold separately for each possible value x of the covariate, with parameters π_{x0} , $f_{x0}(z)$, and $f_{x1}(z)$, yielding

$$f_x(z) = \pi_{x0}f_{x0}(z) + (1 - \pi_{x0})f_{x1}(z) \quad \text{and} \quad \text{fdr}_x(z) = \pi_{x0}f_{x0}(z)/f_x(z). \quad (10.36)$$

The class of cases with covariate x has a priori probability

$$w(x) = \Pr\{x_i = x\}. \quad (10.37)$$

One can picture model (10.35)–(10.37) as an extended version of Figure 10.2, now with a separate arm originating at N for each value of x .

Suppose we are interested in false discovery rates for some category A of cases, for example voxels in the hippocampus in the DTI brain study. We have available a “relevance function” $\rho_A(x)$ relating cases having covariate x to category A . Formally, we interpret $\rho_A(x)$ as the conditional probability of A given x ,

$$\rho_A(x) = \Pr\{A|x\}. \quad (10.38)$$

For the example of Section 10.1, where A was the front half of the brain, $\rho_A(x)$ equaled one or zero as x was ≥ 50 or < 50 (see Figure 2.4).

Theorem 10.1 generalizes to multiple classes.

Theorem 10.3 *Under model (10.35)–(10.37), $\text{fdr}_A(z)$, the local false discovery rate for category A , is related to $\text{fdr}(z)$, the marginal false discovery rate ignoring x , by*

$$\text{fdr}_A(z) = \text{fdr}(z)R_A(z) \quad \text{where } R_A(z) = \frac{E_0\{\rho_A(x)|z\}}{E\{\rho_A(x)|z\}}, \quad (10.39)$$

with $E_0\{\rho_A(x)|z\}$ the null conditional expectation of $\rho_A(x)$ given z ,

$$E_0\{\rho_A(x)|z\} = E\{\rho_A(x)|z, I = 1\}. \quad (10.40)$$

(Proof given below.)

As in (10.9), Theorem 10.3 extends to $\text{Fdr}_A(\mathcal{Z})$, now with $R_A(\mathcal{Z}) = E_0\{\rho_A(z)|z \in \mathcal{Z}\}/E\{\rho_A(z)|z \in \mathcal{Z}\}$.

Even if category A refers to a specific value of covariate x , say $x = x_0$, we still might wish to let $\rho_A(x)$ fall off smoothly as x moves away from x_0 . This would be in the traditional spirit of regression analysis, where we are “borrowing strength” for estimating the conditional expectations in $R_A(z)$ from nearby values of x . As a side benefit, the estimated false discovery rates $\widehat{\text{fdr}}_{x_0}(z)$ would change smoothly as a function of x_0 .

Exercise 10.6 Show how Theorem 10.3 reduces to Theorem 10.1 when $\rho_A(x)$ equals 1 or 0 for x equal or not equal to x_0 .

There is also a general version of Corollary 10.2.

Corollary 10.4 *If the null density $f_{x0}(z)$ does not depend on x ,*

$$f_{x0}(z) = f_0(z) \quad \text{for all } x \quad (10.41)$$

then

$$\text{fdr}_A(z) = c_0 \text{fdr}(z)/E\{\rho_A(x)|z\} \quad (10.42)$$

where

$$c_0 = E\{\rho_A(x)|I = 1\} = \int \rho_A(x)w(x)\pi_{x0} dx/\pi_0. \quad (10.43)$$

Exercise 10.7 Use (10.39)–(10.40) to verify Corollary 10.4.

One can imagine a grand theory of large-scale testing in which every case (10.35) is related to every other by similarities in both their covariates and z -values. The relevance function theory of this section is only a small step in that direction. Its main advantage lies in bringing regression theory, our most flexible “relevance” methodology, to bear on empirical Bayes testing problems.

Proof of Theorem 10.3 From (10.37) and (10.40), the conditional distribution of x given category A is

$$w(x|A) = \frac{w(x)\rho_A(x)}{\rho_A} \quad \text{where } \rho_A = \int w(x)\rho_A(x) dx = \Pr\{A\}. \quad (10.44)$$

The null probability given A is

$$\pi_{A0} = \int w(x|A)\pi_{x0} dx \quad (10.45)$$

with corresponding null density

$$f_{A0}(z) = \int w(x|A)\pi_{x0}f_{x0}(z) dx/\pi_{A0} \quad (10.46)$$

and marginal density

$$f_A(z) = \int w(x|A)f_x(z) dx. \quad (10.47)$$

Together, (10.44)–(10.46) yield

$$\text{fdr}_A(z) = \frac{\pi_{A0}f_{A0}(z)}{f_A(z)} = \frac{\int \rho_A(x)w(x)\pi_{x0}f_{x0}(z) dx}{\int \rho_A(x)w(x)f_x(z) dx}. \quad (10.48)$$

The overall marginal quantities are

$$f(z) = \int w(x)f_x(z) dx \quad f_0(z) = \int w(x)\pi_{x0}f_{x0}(z) dx/\pi_0 \quad (10.49)$$

with $\pi_0 = \int w(x)\pi_{x0} dx$ and

$$\text{fdr}(z) = \frac{\int w(x)\pi_{x0}f_{x0}(z) dx}{\int w(x)f_x(z) dx}. \quad (10.50)$$

Dividing (10.48) by (10.50) gives

$$\begin{aligned} \frac{\text{fdr}_A(z)}{\text{fdr}(z)} &= \left[\frac{\int \rho_A(x)w(x)\pi_{x0}f_{x0}(z) dx}{\int w(x)\pi_{x0}f_{x0}(z) dx} \right] \bigg/ \left[\frac{\int \rho_A(x)w(x)f_x(z) dx}{\int w(x)f_x(z) dx} \right] \\ &= E_0\{\rho_A(x)|z\}/E\{\rho_A(x)|z\}, \end{aligned} \quad (10.51)$$

using Bayes theorem separately on the numerator and denominator of the top equation, which verifies (10.39). $\square$

10.4 Are Separate Analyses Legitimate?

The example of Section 10.1 split the DTI data into two halves, performing separate false discovery rate analyses on each. Is this a legitimate tactic, or do separate analyses compromise inferential properties such as false discovery rate estimates? The answer will turn out to depend on the two faces of empirical Bayes methodology: separation is fine from a Bayesian viewpoint, but some caution is called for in the frequentist domain.

The Bonferroni FWER control method of Chapter 3 suggests why caution might be necessary. Applied to N simultaneous hypothesis tests, it

rejects the null for those cases having p -values p_i sufficiently small to account for multiplicity,

$$p_i \leq \alpha/N \quad (10.52)$$

where α is the significance level. If we separate the cases into two classes of size $N/2$, rejecting for $p_i \leq \alpha/(N/2)$, we effectively double α . Some adjustment of Bonferroni's method is necessary after separation. Changing α to $\alpha/2$ corrects things here, but more complicated situations, like that suggested by Figure 10.4, require careful thought. False discovery rate methods are more forgiving, often (but not always) requiring no adjustment for separate analysis.

This is certainly true from a Bayesian point of view. Let (X, Z, I) be random variables defined as in (10.35), with X indexing possible separation strata (e.g., x in Figure 10.4), Z for z -values, and I equaling 1 or 0 as a random case is null or non-null. Suppose that for each value $X = x$ we have a set $\mathcal{Z}_x$ with conditional Bayesian false discovery rate (2.13) equaling a target value q ,

$$\text{Fdr}_x\{\mathcal{Z}_x\} = \Pr\{I = 1 | X = x \text{ and } Z \in \mathcal{Z}_x\} = q. \quad (10.53)$$

Then the composite region

$$\mathcal{Z} = \bigcup_x \mathcal{Z}_x \quad (10.54)$$

has unconditional Bayesian Fdr equal to q ,

$$\text{Fdr}(\mathcal{Z}) = \Pr\{I = 1 | Z \in \mathcal{Z}\} = q. \quad (10.55)$$

So separate Fdr level- q control rules can be combined without fear of increasing the overall Fdr level, which is to say that separation is a legitimate tactic here. This might seem obvious, but it is not true for FWER control rules.

Exercise 10.8 Prove (10.55).

Frequentist Fdr properties require more care. It is convenient to work within the Poisson-independence assumptions of Section 2.4: that the number of cases N has Poisson distribution $\text{Poi}(\eta)$ and that the z_i are independent. We suppose there are J possible values x of X , and for each x we have a decision rule that decides “non-null” if

$$X = x \quad \text{and} \quad Z \in \mathcal{Z}_x \quad (10.56)$$

(not requiring condition (10.53)). The unbiased false discovery rate estimate (2.45) for stratum x is

$$\widetilde{\text{Fdr}}_x = e_{x0}/(N_x + 1) \quad (10.57)$$

where N_x is the number of cases (x_i, z_i) satisfying (10.56) and e_{x0} is the null hypothesis expectation of N_x ,

$$e_{x0} = \eta w(x) F_{x0}(\mathcal{Z}_x), \quad (10.58)$$

$F_{x0}(\mathcal{Z}_x) = \Pr\{z \in \mathcal{Z}_x | X = x \text{ and } I = 1\}$.

The modified estimate $\widetilde{\text{Fdr}}_x$ is conditionally unbiased for the expected false discovery proportion in stratum x ,

$$E\{\widetilde{\text{Fdr}}_x | X = x\} = E\{\text{Fdp}_x | X = x\} \quad (10.59)$$

as in Lemma 2.4. The combined estimate, ignoring x ,

$$\widetilde{\text{Fdr}}_{\text{comb}} = \frac{e_{\text{tot},0}}{N_{\text{tot}} + 1} \quad \left[e_{\text{tot},0} = \sum_x e_{x0}, \quad N_{\text{tot}} = \sum_x N_x \right] \quad (10.60)$$

is unbiased for the overall expected false discovery proportion.

Simple algebra relates $\widetilde{\text{Fdr}}_{\text{comb}}$ to a weighted average of the $\widetilde{\text{Fdr}}_x$.

Theorem 10.5 *Let*⁴

$$\tilde{w}(x) = \frac{N_x + 1}{N_{\text{tot}} + J} \quad \text{and} \quad \widetilde{\text{Fdr}}_{(\cdot)} = \sum_x \tilde{w}(x) \widetilde{\text{Fdr}}_x \quad (10.61)$$

where J is the number of strata x . Then

$$\widetilde{\text{Fdr}}_{\text{comb}} = \frac{N_{\text{tot}} + J}{N_{\text{tot}} + 1} \widetilde{\text{Fdr}}_{(\cdot)}. \quad (10.62)$$

The factor

$$\text{Pen} \equiv (N_{\text{tot}} + J)/(N_{\text{tot}} + 1) \quad (10.63)$$

can be thought of as a penalty for separating Fdr estimation into J classes. If all the separate unbiased estimates $\widetilde{\text{Fdr}}_x$ equal q , for example, the combined unbiased estimate $\widetilde{\text{Fdr}}_{\text{comb}}$ takes the greater value $\text{Pen} \cdot q$.

The largest penalty occurs for “single event” situations, say with $N_1 = 1$ and $N_2 = N_3 = \cdots = N_J = 0$, where $\text{Pen} = (J + 1)/2$. In this case a separate small value of $\widetilde{\text{Fdr}}_1$ must be multiplied by a factor roughly proportional to

⁴ The weights $\tilde{w}(x)$ are a regularized version of the empirical weights $\hat{w}(x) = N_x/N_{\text{tot}}$ and can be motivated by a vague Dirichlet prior distribution on the stratum probabilities.

the number J of strata, reminiscent of the Bonferroni bound and its adjusted p -values (3.15).

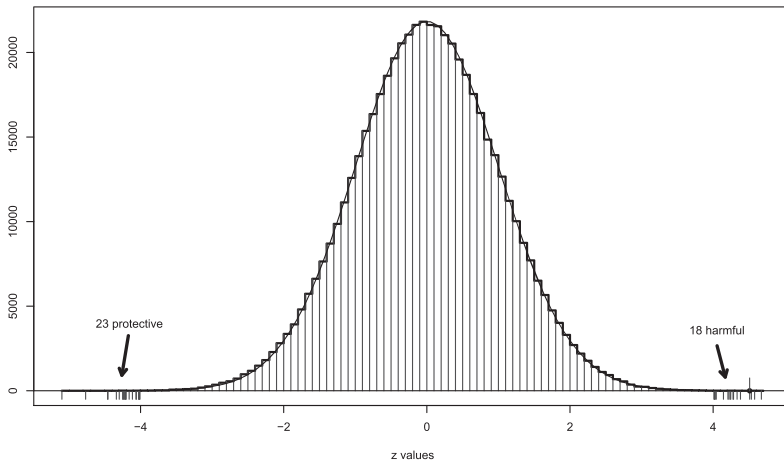


Figure 10.6 Histogram of z -values at $N = 547\,458$ chromosome sites, snp study, comparing 505 cardiovascular patients with 515 controls. Hash marks indicate 41 z -values having $|z_i| \geq 4.0$. Smooth curve is proportional to $\mathcal{N}(0, 1)$ density function, almost perfectly matching the histogram.

Figure 10.6 reports on a snp (*single nucleotide polymorphism*) study examining $N = 547\,458$ chromosome sites on $n = 1020$ subjects, 515 controls and 505 cardiovascular patients. The response variable y_{ij} at site i for person j was the number of observed polymorphisms (i.e., atypical base pairs): $y_{ij} = 0, 1$, or 2 . For each site, a linear model was fit predicting y_{ij} , $j = 1, 2, \dots, 1020$, from disease status (healthy or sick) along with gender and race covariates. A z -value z_i was then computed as in (2.5) from the t -value

$$t_i = \hat{\beta}_i / \widehat{\text{se}}_i \quad (10.64)$$

where $\hat{\beta}_i$ and $\widehat{\text{se}}_i$ were the usual regression coefficient and its standard error for disease status.

The histogram of the 547 458 z -values is almost perfectly matched by a $\mathcal{N}(0, 1)$ density. In this example the theoretical null (2.6) seems trustworthy, and the null probability π_0 (2.7) must be nearly 1. Only 41 of the snps give z -values exceeding 4 in absolute value, 18 on the right and 23 on the left. Table 10.2 shows the 41 z -values, their chromosome, and the

number of sample sites per chromosome. For instance, chromosome 9 with $N(9) = 26\,874$ sites had only one z -value making the list, at $z_i = 4.51$.

Table 10.2 *The 41 snps with $|z_i| \geq 4$, snp study, Figure 10.6, showing chromosome, number of measured snps per chromosome, z -value, unbiased Fdr estimate $\widehat{\text{Fdr}}_x$ (10.57), penalized version $\widehat{\text{Fdr}}_x^+$ (10.66), combined q -value $\hat{q}_{\text{comb}}$ (10.67), and separate-chromosome q -value $\hat{q}_{\text{sep}}$, as in (10.68).*

Chrom x	# snps	z -value	$\widehat{\text{Fdr}}_x$	$\widehat{\text{Fdr}}_x^+$	$\hat{q}_{\text{comb}}$	$\hat{q}_{\text{sep}}$
1	42 075	-4.21	.53	.80	.57	.27
1	42 075	4.38	.53	.80	.72	.50
1	42 075	-4.22	.53	.80	.62	.34
1	42 075	-4.06	.53	.80	.88	1.00
2	45 432	-4.16	.72	1.00	.65	.72
2	45 432	-4.02	.72	1.00	.90	1.00
2	45 432	-4.46	.72	1.00	.64	.12
5	34 649	-4.21	1.00	1.00	.60	.88
6	36 689	4.25	.58	.87	.74	.39
6	36 689	4.14	.58	.87	.67	1.00
6	36 689	-4.34	.58	.87	.79	.17
7	30 170	4.28	.48	.72	.74	.19
7	30 170	-4.23	.48	.72	.64	.35
7	30 170	-4.06	.48	.72	.87	1.00
8	31 880	4.33	.67	1.00	.73	.48
8	31 880	4.53	.67	1.00	.63	.09
9	26 874	4.51	.85	1.00	.59	.17
10	29 242	-4.01	.93	1.00	.84	1.00
11	27 272	4.22	.43	.65	.63	.22
11	27 272	-4.77	.43	.65	.51	.05
11	27 272	-4.02	.43	.65	.84	.79
12	27 143	4.01	.34	.52	.83	1.00
12	27 143	4.58	.34	.52	.64	.04
12	27 143	-4.11	.34	.52	.74	.54
12	27 143	-5.10	.34	.52	.19	.00
13	20 914	-4.01	.33	.50	.81	1.00
13	20 914	-4.20	.33	.50	.56	.19
13	20 914	-4.03	.33	.50	.91	.58
15	16 625	4.03	.26	.39	.93	.93
15	16 625	-4.46	.26	.39	.57	.05
15	16 625	-4.23	.26	.39	.66	.19
17	14 341	-4.25	.30	.45	.78	.15
17	14 341	-4.24	.30	.45	.70	.32
18	16 897	4.24	.27	.40	.68	.19
18	16 897	4.02	.27	.40	.87	.98
18	16 897	4.67	.27	.40	.54	.02
19	9501	4.04	.20	.30	.90	.51
19	9501	4.21	.20	.30	.59	.12
20	14 269	4.28	.30	.45	.77	.27
20	14 269	-4.30	.30	.45	.79	.12
21	8251	4.02	.26	.39	.88	.48

As an example of Theorem 10.5, suppose we let $x = 1, 2, \dots, 22$ index

chromosomes (chromosome 23 was excluded from the study) and take

$$\mathcal{Z}_x = \{|z| \geq 4\} \quad \text{for } x = 1, 2, \dots, 22. \quad (10.65)$$

In this case $N_{\text{tot}} = 41$, $J = 22$, and $\text{Pen} = (41 + 22)/(41 + 1) = 1.50$. Columns 4 and 5 of Table 10.2 show $\widehat{\text{Fdr}}_x$ and the penalized value

$$\widehat{\text{Fdr}}_x^+ = \text{Pen} \cdot \widehat{\text{Fdr}}_x. \quad (10.66)$$

The idea behind $\widehat{\text{Fdr}}_x^+$ is to adjust the separate estimate $\widehat{\text{Fdr}}_x$ so that the adjusted average equals the combined estimates $\widehat{\text{Fdr}}_{\text{comb}}$ (10.62). None of the separate estimates are strikingly small, though chromosome 19 gets down to $\widehat{\text{Fdr}}_{19} = 0.20$, adjusted to $\widehat{\text{Fdr}}_{19}^+ = 0.30$.

Exercise 10.9 Verify $\widehat{\text{Fdr}}_{19} = 0.20$.

The sixth column of Table 10.2 concerns application of the Benjamini–Hochberg FDR control algorithm to the combined set of all 547 458 snps. Tabled is $\hat{q}_{\text{comb},i}$, the minimum value of q for which BH(q) would declare z_i non-null. Ordering the z_i values according to absolute value, $|z_{(1)}| > |z_{(2)}| > \dots$, the q -value corresponding to the i th ordered case is

$$\hat{q}_{\text{comb},(i)} = 2 \cdot N \cdot \Phi(|z_{(i)}|)/i. \quad (10.67)$$

Only the most extreme case, $z_i = -5.10$, has even a moderately interesting result, with $\hat{q}_{\text{comb}} = 0.19$. (Note that (10.67) is based on two-sided p -values.)

Exercise 10.10 Justify (10.67) in terms of Corollary 4.2 (4.30).

By contrast, q -values calculated separately for the different chromosomes yield several interestingly small values $\hat{q}_{\text{sep},i}$, the last column in Table 10.2: $z_i = -4.51$ on chromosome 9, for instance, has

$$\hat{q}_{\text{sep}} = 2 \cdot 26874 \cdot \Phi(-4.51)/1 = 0.17. \quad (10.68)$$

The histogram of the 26 874 chromosome 9 z -values looks just like Figure 10.2 except with 4.51 as the only outlier (indicated by the dot and bar at the far right of Figure 10.6), giving a strong visual impression of a significant discovery. Sixteen of the 41 snps have $\hat{q}_{\text{sep}} \leq 0.20$, with six of them ≤ 0.10 .

Can these much more optimistic results be trusted? Probably not, at least not from the point of view of Theorem 10.5. Changing 4 in (10.65) to 4.5 increases the penalty factor to $\text{Pen} = 5.2$, weakening any interest in the chromosome 9 outcome. We do not have a neat formula like (10.66) for

adjusting the q -values, but if we have been cherry-picking results from all 22 chromosomes then some sort of adjustment must be called for.

The snp study raises several points of concern relating to false discovery rate analysis, especially as carried out at the extremes of a data set:

- In actuality, the data for chromosome 9 was analyzed first, at which point 4.51 looked like a clear discovery. The problem here, as with all frequentist methods, lies in defining an appropriate frame of reference. Whether mathematical statistics can improve on what Miller lamented as “subjective judgement” remains an open question, though perhaps the relevance considerations of the previous section offer some guidance.
- The variability of $\overline{\text{Fdr}}(\mathcal{Z})$ or $\widetilde{\text{Fdr}}(\mathcal{Z})$ as an estimate of the Bayes probability $\text{Fdr}(\mathcal{Z})$ is enormous at the extremes of the z -values, where the crucial quantity $e_+(\mathcal{Z})$ in (2.39) may be 1 or less. This undercuts faith in the Bayesian interpretation of our Fdr estimates (which, in (10.53)–(10.55), supported the legitimacy of separate analyses). Fdr and FWER methods coalesce at the extremes, both requiring frequentist caution in their use.
- These concerns vanish as we move in from the extremes of the z -value histogram. The separation penalty factor Pen (10.63) decreases to 1 when N_{tot} is much larger than J , while the implication of large $e_+(\mathcal{Z})$ supports Bayesian interpretation of our Fdr estimates.
- Local fdr methods are of no help in the extreme tails, where semi-parametric fitting methods for $f(z)$, Section 5.2, cannot be trusted.
- There is a certain “angels on the head of a pin” quality to the Fdr results in Table 10.2. They require belief in calculations such as

$$\Pr_0\{z_i \geq 4\} = \Phi(-4) = 3.2 \cdot 10^{-5}, \quad (10.69)$$

well beyond the range of accuracy of the normal-theory approximations for statistics (2.5), (10.64). Such beliefs are necessary for any multiple testing procedure applied to the snp data.

- If all $N = 547\,458$ snps were null, we would expect 34.7 z -values exceeding 4 in absolute value (according to the two-sided version of (10.69)). The “higher criticism” test (4.42) rejects this hypothesis at the 0.05 level for $N_{\text{tot}} \geq 45$ (4.43), well beyond the observed value 41.
- Searching for enriched classes of snps is a promising tactic. However, in this case grouping snps from the same gene and applying the slope test (10.26) yielded no new discoveries.

10.5 Comparability

Empirical Bayes methods involve each case learning from the experience of others: other baseball players, other genes, other policemen, other snps. To make this believable, the “others” have to be similar in nature to the case at hand, or at least not obviously dissimilar. We wouldn’t try to learn about prostate cancer genes from a leukemia study, even if the z -value histograms resembled each other. *Comparability*, our subject here, refers to questions of fair comparison in large-scale studies. It relates to the relevance calculations of Section 10.3, but with more emphasis on purely statistical corrections and less on the underlying covariate connections.

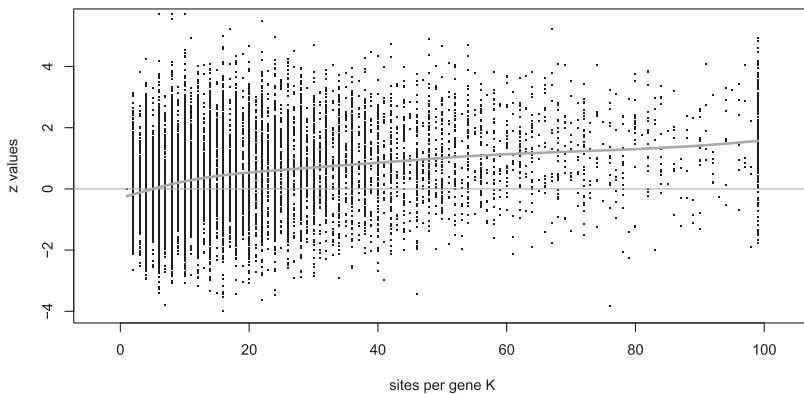


Figure 10.7a z -values for chi-square data, Section 6.1B, plotted versus number K of binding sites for each gene. Smoothing spline regression (heavy curve) shows that sites with large K tend toward larger z -values.

Figure 10.7a and Figure 10.7b illustrate two examples. The first refers to the chi-square data of Figure 6.1b. Here the $N = 16882$ z -values are plotted versus K , the number of binding sites within each gene. We see that the z -value algorithm of Section 6.1B tended to produce larger values for larger K . Recentering the z -values by subtraction of the regression curve produces a fairer comparison. (Repeating the $\widehat{\text{fdr}}$ analysis of Figure 6.1b after recentering, the seven z -values on the extreme right no longer had $\widehat{\text{fdr}} < 0.2$.)

A more egregious example appears in Figure 10.7b. Thirteen brain tumor patients were measured on two microarrays each, one from the tumor center, the other from its periphery. One-sample t -statistics with 12 degrees

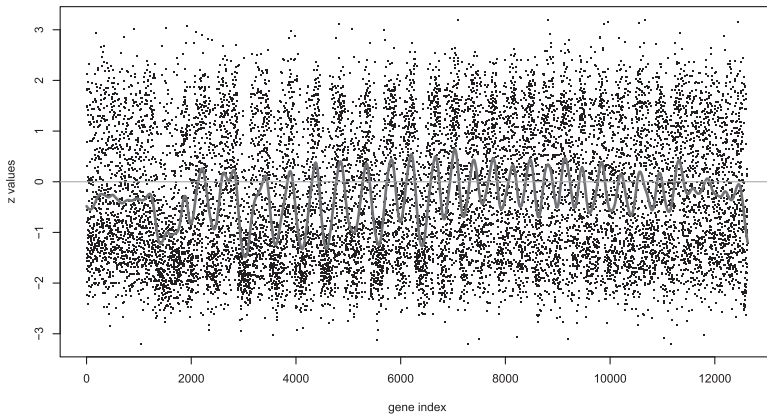


Figure 10.7b z -values for $N = 12\,625$ genes, $n = 13$ patients, brain tumor study, plotted versus their order of being recorded. Smoothing spline (heavy curve) reveals a strong periodic disturbance.

of freedom, based on center/periphery differences, were calculated for each of $N = 12\,625$ genes and converted to z -values as in (2.5). The z_i values are plotted versus i , the order in which they were read. A substantial periodic disturbance is evident, probably caused by defects in the microarray reading mechanism.⁵

Subtracting off the oscillating curve is a minimal first step toward improved comparability across genes. It may not be enough. In the DTI example of Figure 10.4, the *variability* of the z -values changes along with their running median. Application of Corollary 10.2, as in Figure 10.5, offers a more forceful equilibration of the z -values.

Questions of comparability can arise more subtly. The early microarray literature focused on “fold-change” rather than t -statistics in two-sample comparisons. If the expression-level entries x_{ij} are logarithms of the original measurements (as with the prostate data), this amounts to analyzing the numerators of the t -statistic (2.2),

$$d_i = \bar{x}_i(2) - \bar{x}_i(1) \quad (10.70)$$

rather than $t_i = d_i/s_i$.

A recent experimental comparison of microarray platforms used *spike-in*

⁵ The period matches the width of the rectangular microarray chip from which they were read.

samples⁶ to test agreement among different commercial technologies. The good news was that the different microarray types agreed in locating non-null genes. Less good was the authors' recommendation of fold-change rather than t -statistics for testing purposes, on the grounds that the d_i gave more stable results.

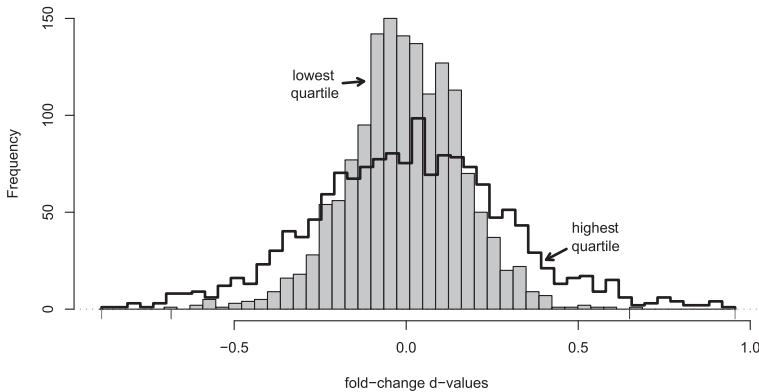


Figure 10.8 Fold-change statistics d_i (10.70) for the prostate data of Section 2.1. The genes have been divided into quartiles on the basis of s_i , the estimated standard error. The d_i histogram for the highest s_i quartile is much more dispersed than that for the lowest quartile.

This last statement almost has to be true: $t_i = d_i/s_i$ is usually noisier than d_i because of the randomness in the denominator. Stability is less important than comparability though, as illustrated in Figure 10.8. There the prostate data of Section 2.1 has been reanalyzed in terms of fold-changes d_i . The $N = 6033$ genes have been divided into quartiles according to the size of s_i , the estimated standard error. Histograms show d_i distributions in the lowest and highest quartiles. Not surprisingly, the histogram is much wider for the highest quartile.

Exercise 10.11 Why isn't this surprising? *Hint:* Consider the situation $x_{ij} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \sigma_i^2)$.

A false discovery rate analysis based on the d_i values would preferentially select genes i with high variability in their expression levels x_{ij} , which

⁶ Artificially constructed biological samples where it is known which genes are null or non-null.

is probably irrelevant to their null/non-null status. The obvious way to correct for this is to divide d_i by s_i , getting us back to t -statistics t_i . (The corresponding figure for the t_i values shows no difference between the two histograms.)

The popular SAM algorithm described in Section 4.5 advocates use of a compromise test statistic,

$$u_i = \frac{d_i}{a_0 + s_i}. \quad (10.71)$$

Choosing $a_0 = 0$ makes $u_i = t_i$, while letting a_0 approach infinity makes u_i equivalent to d_i for testing purposes. As a_0 increases, stability improves (especially for genes that have very small values of s_i) but at the cost of decreased comparability. With a_0 equal the median of the N s_i values, one of the SAM choices, the equivalent of Figure 10.8 still shows a large discrepancy between the two histograms.

Double standardization (8.2) improves comparability across cases. It makes fold-changes d_i equivalent to t -statistics. Full double standardization isn't a necessity, but if the rows of X are of vastly different scales, non-comparability may be a concern, particularly for more complicated situations than two-sample testing.

Empirical Bayes methods require the pooling of information across sets of, hopefully, similar cases. This chapter's topic, combination and separation, relevance, and comparability, concerned when and how the pooling should be done. The discussion raised more questions than it settled, but they are good questions that deserve some thought before embarking on any large-scale data analysis.

Notes

Most of the material in Section 10.1 through Section 10.4 originated in Efron (2008b). Genovese et al. (2006) consider more qualitative situations than those of Section 10.1, where A or B might be classes of greater or less a priori null probability. Their "weighted BH" rule transforms z_i into values z_{Ai} or z_{Bi} depending on the class and then carries out a combined Fdr analysis, rather than keeping z_i the same but using different Fdr _{A} or Fdr _{B} analyses. Ferkingstad et al. (2008) explore the dependence of Fdr(z) on x by means of explicit parametric models.

Q values, Section 10.4, were introduced by Storey (2002). Efron and Morris (1971, 1972) used relevance functions for James–Stein estimation where, as in Section 10.3, the question was one of combining data from

more or less similar situations. The snp study, ongoing, comes from the Stanford laboratory of T. Quertermous. Figure 10.7b is based on an unpublished study from the Stanford School of Medicine.